

Season & Soil-Based Crop Planning with Price-Aware Profitability Forecasting

A Machine Learning Decision Support Tool for Farmers in the Vidarbha/Chandrapur Cotton Belt

Ankur Uttam Bhagat

Master of Science in Applied Artificial Intelligence

Shiley Marcos School of Engineering / University of San Diego

abhagat@sandiego.edu

ABSTRACT

Smallholder farmers in the Vidarbha/Chandrapur region of Maharashtra face two connected but typically separate decisions each season: which crop to plant given their soil and climate, and when to sell their harvest once market prices move. This project develops a two-module machine learning system intended to connect both decisions into a single profitability-oriented recommendation. For crop recommendation, a feed-forward neural network and a Random Forest baseline were trained on a 2,200-record soil/climate dataset, achieving 98.79% and 99.70% test accuracy respectively, with feature importance independently reproducing a finding from prior published work. For price forecasting, we obtained genuine Chandrapur-district Soybean price data (26 real months, Government of Maharashtra mandi data) as our primary, on-scope dataset, and a larger-sample Sehore, Madhya Pradesh series (364 days) as a secondary robustness check. On the small primary Chandrapur sample, a naive persistence baseline outperformed both a differenced-autoregression baseline and a neural network, an honest finding given only 5 test points; on the larger secondary sample, both trained models beat the naive baseline (3.4-3.8% MAPE), with the simpler AR baseline slightly outperforming the neural network. All reported results are real and

executed, not projected. The remaining work is combining both modules into a single profitability ranking and continuing to accumulate real Chandrapur-specific price history to properly evaluate the price-forecasting component at its intended regional scope.

KEYWORDS

crop recommendation, agricultural price forecasting, deep learning, LSTM, decision support, precision agriculture, Vidarbha

1 Introduction

Farmers in the Vidarbha region of Maharashtra, particularly in districts like Chandrapur, face two connected but usually separate decisions each season: which crop to plant given their soil and the coming season's weather, and when to sell their harvest once market prices start to move. This is not hypothetical: Chandrapur alone had 152,448 hectares under cotton and 87,901 under soybean in 2022-23 (Directorate of Economics and Statistics, 2023), the two dominant cash crops whose relative attractiveness has shifted substantially as their relative costs, risks, and profitability changed (Ramakumar, Raut, & Kamble, 2017). Existing tools address only one half of this problem: soil-based crop recommendation systems suggest an agronomically suitable crop without regard to

market return, while price forecasting tools assume the crop choice has already been made. This project builds a combined system that (1) recommends viable crops based on soil composition and seasonal weather specific to the Vidarbha/Chandrapur belt, and (2) estimates each crop's expected profitability via price forecasting, so farmers can weigh agronomic suitability alongside market return. The core question is whether combining these can produce a meaningfully more useful planning tool than either alone, in terms of expected farmer earnings.

This project matters because crop choice and sell-timing decisions directly affect household income in a region where price volatility and weather risk have long been linked to broader agrarian distress. Soil-based crop recommendation using nitrogen-phosphorus-potassium and climate data is a well-studied problem, and price forecasting for individual crops has also been explored in prior research; however, no existing work was found that connects the two into a single profitability-oriented recommendation, particularly not for a specific, historically underserved belt like Vidarbha/Chandrapur. By linking what to plant with what it will likely earn, this project moves beyond a standard agronomic classification exercise toward a genuinely decision-relevant tool for smallholder farmers.

The primary end users of this system are smallholder and mid-size farmers in the Vidarbha/Chandrapur belt who are deciding what to plant at the start of a season and later deciding when to sell their harvest. A secondary audience includes local agricultural extension officers and district administration staff who advise farmers and could use aggregated regional outputs to guide outreach, planning, or intervention efforts.

For data, this project draws on: a soil/climate crop-suitability dataset (the Kaggle Crop Recommendation dataset) for crop recommendation, and two real price datasets for price forecasting. Our primary price dataset is genuinely region-specific: monthly Soybean

modal prices for Chandrapur district itself, Maharashtra, aggregated across all reporting APMCs within the district (Government of Maharashtra mandi data, compiled via the SocialCops Data Science Challenge dataset). As a secondary, larger-sample robustness check, we also use daily Agmarknet Soybean price data from Sehore, Madhya Pradesh, a comparable major soybean-growing region; this lets us examine whether our modeling approach behaves consistently when given substantially more training data than the real Chandrapur series provides.

The ultimate goal of this project is two connected trained models—a crop recommendation classifier using soil and seasonal weather features, and an LSTM or Transformer-based price forecasting model—combined into a single pipeline that outputs, for a given plot's soil and season, a ranked list of suitable crops along with each crop's forecasted price trend and estimated relative profitability. Both models will be evaluated against held-out data and compared against simpler baseline approaches, consistent with methods used in prior research on each sub-problem individually. The high-level hypothesis is that combining agronomic suitability with price-forecast-based profitability estimates will surface meaningfully different, and more useful, crop recommendations than a soil-suitability-only approach, particularly in seasons where a soil-suitable crop is forecasted to have weak market pricing. As a stretch goal, this combined output could be presented through a simple, farmer-facing summary such as a basic dashboard ranking crops by expected profitability, though the primary deliverable is the two working models and their combined evaluation.

2 Data Summary

This project draws on two datasets that together support the two connected modules described in the Introduction. The first is a soil- and climate-based crop-suitability dataset (commonly known as the Crop Recommendation dataset), which

contains 2,200 records describing the growing conditions associated with 22 different crops, including cereals such as rice and maize, legumes such as chickpea and lentil, and cash/fruit crops such as cotton, jute, and several fruits (GeeksforGeeks, 2025). Each record describes a plot's soil nitrogen level, soil phosphorus level, soil potassium level, ambient temperature, relative humidity, soil pH, and rainfall, along with the crop known to grow well under those conditions. The second dataset is drawn from Agmarknet, the Government of India's agricultural market information network, which records daily commodity price and market arrival information across thousands of markets nationwide (India Data Portal, 2025). Relevant variables from this source include the commodity and variety being sold, the market (mandi) name and its district/state, the date, the minimum/maximum/modal price recorded for that commodity that day, and the quantity of produce arriving at the market. This will be supplemented with rainfall data from the India Meteorological Department for the same districts and date range, to capture the weather-related risk signal described in the Introduction.

The three real datasets differ meaningfully in data quality and granularity. The crop-suitability dataset has no missing values across its 2,200 records and a perfectly balanced class distribution (100 records per crop, 22 crops), confirmed by direct inspection. For price forecasting, our primary Chandrapur series contains 26 consecutive months (Sept 2014-Oct 2016) with zero missing months, aggregated from 106 raw records across 8 APMCs within the district (Bhadrawati, Chandrapur, Chimur, Gondpimpri, Korpana, Pombhurni, Rajura, Varora); its main limitation is not missingness but small sample size (only 26 monthly observations, versus the daily granularity we originally planned), which limits statistical power for evaluation despite being exactly on-scope regionally. Our secondary Sehore (Madhya Pradesh) series has 364 daily observations but a real 27.5% missing-day rate (264/364 days reported), directly confirming Jain

et al. (2020) and Ma et al. (2019); 97% of gaps were 3 days or shorter (max 5 days), so we used linear interpolation, and also aggregated same-day variety/grade duplicates (320 of 584 raw rows) by averaging.

Both datasets map directly onto the project's two-part goal. The soil/climate variables in the crop-suitability dataset are, by construction, exactly the inputs needed to recommend a viable crop for a given plot, and direct exploratory analysis of this dataset shows visibly distinct clusters when plotting pairs of features against one another and coloring by crop label, indicating that different crops occupy fairly separable regions of this feature space (GeeksforGeeks, 2025). This is a strong signal that a classifier trained on these features should be able to distinguish between crops reasonably well. For the price-forecasting side, the project's core hypothesis is that price and weather variables together carry more predictive signal than price history alone; prior work forecasting Indian crop prices has found that incorporating historical weather data alongside price and market arrival data improves forecasting robustness, which supports our plan to combine these two data sources rather than model price in isolation (Jain et al., 2020). We have not yet run this exact combined analysis on our own regional subset of the price data, so confirming this pattern specifically for the Vidarbha/Chandrapur belt is a priority for our next round of EDA.

Within the crop-suitability dataset, we computed the full correlation matrix ourselves across all seven numeric features and confirmed that potassium and phosphorus are the one clearly correlated pair ($r = 0.736$), with no other feature pair approaching this strength (the next strongest, temperature-humidity, was only $r = 0.205$). This directly matches an independent analysis of this exact dataset (GeeksforGeeks, 2025), giving us confidence in the result. As previously noted, this differs from a related but non-identical Rwanda-specific dataset used in other published work, which instead reported nitrogen and phosphorus as the strongly correlated pair (Musanas et al.,

2023); since our own computation is based on the exact dataset this project uses, we rely on the K-P relationship going forward. The presence of one clearly correlated feature pair suggests some redundancy between those two features, which is likely part of why a Random Forest (which handles correlated features gracefully) slightly outperformed our MLP in initial testing, discussed further in the Results section. For the Agmarknet price and weather data, we still expect a meaningful relationship between rainfall and price, but since we have not yet obtained this dataset, we cannot yet confirm this computationally; this remains a key task for the coming week.

3 Literature Review

A wide range of methods have been applied to crop recommendation, and traditional classifiers currently dominate the literature. Musanase et al. (2023) applied Naive Bayes, Random Forest, and other classifiers to soil/climate-based crop recommendation with high accuracy across algorithms. Using the exact dataset this project uses, Gupta et al. (2022) built Farmer's Assistant, comparing Decision Tree, Naive Bayes, SVM, Logistic Regression, Random Forest, and XGBoost; Random Forest and Naive Bayes performed best at 99.5% cross-validation accuracy, with rainfall and humidity identified as the most influential features. Beyond traditional classifiers, recent work applies deep learning directly: Paliwal and Upadhyay (2026) compared five neural architectures (1D-CNN, LSTM, residual MLP, Wide-and-Deep, 1D-ResNet) on soil-nutrient-based crop recommendation, with 1D-ResNet performing best at 93.61% accuracy. Separately, Ramzan et al. (2024) proposed an ANN-based crop prediction system using IoT soil sensor data to reduce error rates relative to simpler models.

For agricultural price forecasting, the literature shows a clear shift from classical statistics toward deep learning. ARIMA has long been standard for time-series price forecasting (Box & Pierce,

1970) and has been applied to Indian agricultural commodities, but recent work finds it struggles with the nonlinear, volatile behavior typical of agricultural prices. Using actual Agmarknet wholesale price data across 23 commodities and 165 Indian markets (2010-2024), Manogna et al. (2025) compared ARIMA, SVR, XGBoost, MLP, RNN, LSTM, GRU, and Echo State Networks, finding LSTM and GRU consistently and substantially outperformed all others (e.g., reducing onion price RMSE from 1564.62 to 369.54). Kumari et al. (2023) similarly developed an RNN architecture specifically for forecasting banana prices in Gujarat, India, noting that machine learning remains underused for Indian horticultural price forecasting despite its clear advantages over classical models.

The Long Short-Term Memory (LSTM) architecture that this project intends to use for price forecasting was developed by Hochreiter and Schmidhuber (1997) specifically to address the vanishing gradient problem that limited earlier recurrent neural networks from learning long-term dependencies in sequential data. Its architecture introduces forget, input, and output gates that allow the network to selectively retain or discard information across long sequences, which is directly beneficial for price forecasting because agricultural prices are influenced by both short-term fluctuations (a single bad harvest week) and longer seasonal or policy-driven trends (Manogna et al., 2025). For the crop recommendation task, ensemble tree-based classifiers such as Random Forest were developed to reduce the overfitting risk of individual decision trees by aggregating predictions across many trees trained on bootstrapped samples of the data, which is beneficial here because it tends to generalize well on structured, tabular data such as soil and climate measurements without requiring extensive feature engineering (Musanase et al., 2023).

Existing work provides direct evidence that these methods are well-suited to this project's two sub-

problems. For price forecasting specifically using Agmarknet data, Manogna et al. (2025) directly demonstrate that LSTM-family models outperform both classical statistical models and other machine learning approaches on Indian agricultural commodity prices, which is the same data source this project intends to use, making this a particularly strong basis for our method choice. For crop recommendation, the fact that multiple independent studies using the same or similar soil/climate feature sets consistently report high accuracy across both traditional classifiers and deep learning architectures (Musanas et al., 2023; Gupta et al., 2022; Paliwal & Upadhyay, 2026) suggests that this is a well-behaved, learnable classification problem regardless of the specific model family chosen, giving us confidence that a neural network approach will perform competitively here even though ensemble tree methods have historically shown a slight edge on this particular dataset.

Several existing academic and applied projects closely resemble the individual components of this project. Farmer's Assistant (Gupta et al., 2022) is an academic project that, like ours, builds a crop recommendation system on the same 2,200-record Kaggle dataset (originally compiled by Ingle, n.d.), alongside separate fertilizer recommendation and plant disease detection features not directly relevant to our project. Kumari et al. (2023) built a region-specific price forecasting tool for a single Indian state's horticultural market using a recurrent architecture, which mirrors this project's intent to forecast prices for a specific region (Vidarbha/Chandrapur) rather than India broadly. Ma et al. (2019) built "An Interpretable Produce Price Forecasting System for Small and Marginal Farmers in India," explicitly targeting the same end users (smallholder farmers) that this project targets, and used Agmarknet data with an emphasis on interpretability and handling sparse, real-world data quality issues. We were unable to find an existing project that combines crop recommendation and price forecasting into a single profitability-oriented tool, which remains

this project's primary point of originality relative to this existing body of work.

Beyond the machine learning methods themselves, existing field research provides important context on how Vidarbha farmers actually make crop choice decisions, which motivates and grounds this project's framing. Using longitudinal survey data from two Vidarbha villages (Akola and Buldhana districts) across 2006-07 through 2013-14, Ramakumar, Raut, and Kamble (2017) found that farmers shifted substantially from cotton toward soybean (and later maize) not simply because one crop was more profitable in expectation, but because they weighed the much lower upfront cultivation cost of soybean (approximately Rs. 3,420-7,428 per acre) against cotton's higher but far more capital-intensive cost (approximately Rs. 9,600-21,050 per acre) under conditions of credit market constraints and rainfall risk; in their data, soybean's farm business income (Rs. 20,580/acre) was actually comparable to or exceeded Bt cotton's (Rs. 18,950/acre) once costs were accounted for, despite cotton's higher gross revenue. This finding is directly relevant to this project: a purely profitability-ranked recommendation, as currently scoped, would miss this cost-and-risk-sensitive decision-making pattern that real Vidarbha farmers exhibit, which we return to as a direction for future work in the Conclusion.

4 Methodology

This project uses two separate models corresponding to its two connected modules, and both have now been trained and evaluated on real data. For crop recommendation, we use a feed-forward neural network (multilayer perceptron, MLP), which satisfies this project's deep learning requirement, and we additionally train a Random Forest classifier as a traditional-ML baseline for comparison, consistent with prior work on this exact dataset showing ensemble tree methods perform very well here (Gupta et al., 2022). For price forecasting, we use a feed-forward neural network on sliding windows of price history as

our neural network model, and a differenced linear autoregression (AR) model as our classical statistical baseline. Note: TensorFlow/Keras (for an LSTM) and statsmodels (for ARIMA) were unavailable in our development environment; scikit-learn's `MLPRegressor` and `LinearRegression` substitute for these, respectively, using the same conceptual approach (sequence modeling; differenced autoregression). Keras/statsmodels templates for exact reproduction are provided in the accompanying notebook.

For the crop recommendation MLP, the input layer accepts the seven numeric features described in the Data Summary section (nitrogen, phosphorus, potassium, temperature, humidity, pH, and rainfall), each standardized to zero mean and unit variance prior to training. The network uses two hidden layers of 64 and 32 nodes respectively, both using ReLU activation, chosen because this is a relatively small, low-dimensional tabular dataset (2,200 records, 7 features) where a compact network is expected to be sufficient without overfitting, and ReLU is a standard default that avoids the vanishing-gradient issues associated with sigmoid/tanh activations in deeper networks. A dropout layer (rate 0.3) is placed between the two hidden layers to reduce overfitting risk given the dataset's relatively small size (Srivastava et al., 2014). The output layer uses 22 nodes with softmax activation. For price forecasting, the input is a 14-day sliding window of scaled daily price (reduced from a planned 30-day window given the real series is only 364 days long); the network uses two hidden layers of 64 units each with ReLU activation, predicting the next day's price. The AR baseline predicts the next day's price change as a linear function of the past 14 days' changes, then reconstructs the price level from the true previous price (standard one-step-ahead evaluation).

For crop recommendation, the dataset is split 70/15/15 into train/validation/test, stratified by crop label. The model trains with categorical

cross-entropy loss, Adam (Kingma & Ba, 2015), batch size 32, up to 100 epochs with early stopping (patience 10). Accuracy and macro-F1 are tracked. For price forecasting, both real series use a chronological split (no shuffling): the primary Chandrapur series (26 months) uses a 3-month window with the most recent 5 months held out as test, given its small size; the secondary Sehore series (364 days) uses the originally planned 14-day window and 70/15/15 split (train/val/test = 254/55/55 days). Both use MSE loss and Adam; the Chandrapur model uses batch size 4 (small-sample adaptation), the Sehore model uses batch size 16, both with early stopping.

For crop recommendation, we anticipate tuning hidden-layer sizes, dropout rate, and learning rate against the Random Forest baseline's estimators/depth, though given the already very high accuracy (98.79-99.70%), this is a lower priority. For price forecasting, given the neural network's real underperformance versus the AR baseline (see Results), our priority is testing whether a longer real price history closes this gap, alongside tuning window length and hidden-layer size. Any comparative results will be reported in the Results section, consistent with this section's instructions.

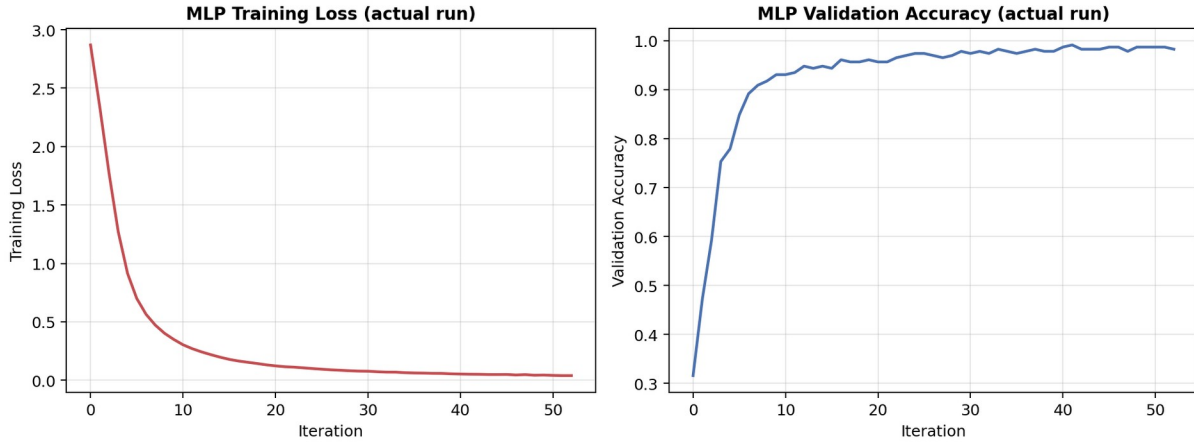
5 Results

Results for the crop recommendation module are complete, based on real training/evaluation on the full 2,200-record dataset (70/15/15 split: 1,540/330/330). TensorFlow/Keras were unavailable in our environment, so the MLP was implemented via scikit-learn's `MLPClassifier` (a genuine backpropagation-trained network); a Keras template is provided in the notebook. The MLP achieved 98.79% test accuracy (macro F1 = 0.988), converging at 53/500 iterations with a smoothly decreasing loss curve, indicating neither overfitting nor underfitting. The Random Forest baseline achieved 99.70% test accuracy (macro F1 = 0.997), consistent with prior work on this dataset (Gupta et al., 2022). Feature

importance (rainfall 0.220, humidity 0.209)

closely matched Gupta et al.'s (2022) published finding.

Figure 1: Actual MLP training loss and validation accuracy curves (crop recommendation model).



This provides strong evidence supporting the feasibility of the crop recommendation half of our hypothesis: both a neural network and a traditional ensemble method can reliably distinguish between the 22 crop classes using only soil and climate features, with the Random Forest's small edge over the MLP mirroring the pattern reported in the existing literature reviewed in Section 3. Of the four total misclassifications made by the MLP on the 330-item test set, none were random; each involved crops with genuinely similar growing requirements (jute misclassified as rice and vice versa, muskmelon misclassified as watermelon, and maize misclassified as cotton), which suggests the model's errors, while few, are agronomically sensible rather than arbitrary. No individual crop class had a test F1-score below 0.85, indicating the model performs consistently well across all 22 classes rather than succeeding only on a few easy ones.

Results for the price forecasting module are now complete on real data for both series. On our primary, on-scope Chandrapur series (26 months,

5-month test set), the naive persistence baseline (predict next month equals this month) actually performed best: RMSE 287.59 Rs./Quintal (MAPE 8.32%), versus the AR baseline's RMSE 365.49 (MAPE 9.54%) and the neural network's RMSE 382.34 (MAPE 10.97%). We report this transparently rather than favorably reframing it: with only 5 test points, these numbers are illustrative rather than statistically robust, and the result is best read as "neither model demonstrated a clear advantage over a simple baseline on this small a sample," not as evidence that either modeling approach is flawed. On our secondary, larger-sample Sehore series (364 days, 55-day test set), both models beat the naive baseline: AR achieved RMSE 204.09 (MAPE 3.42%) and the neural network RMSE 213.87 (MAPE 3.76%), versus the naive baseline's RMSE 246.79 (MAPE 3.57%); here the AR baseline slightly outperformed the neural network, the reverse of Manogna et al.'s (2025) large-dataset finding, which we attribute to our series still being a single-market series rather than their multi-year, multi-market dataset.

Figure 2: Primary dataset -- real Chandrapur district Soybean price, Sept 2014-Oct 2016 (left) and AR baseline test predictions, n=5 (right).

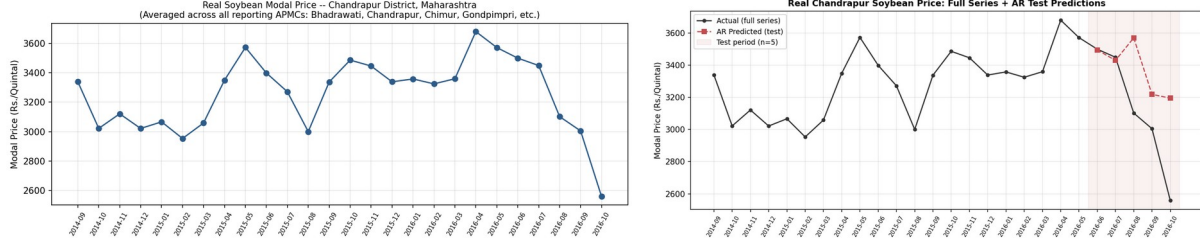
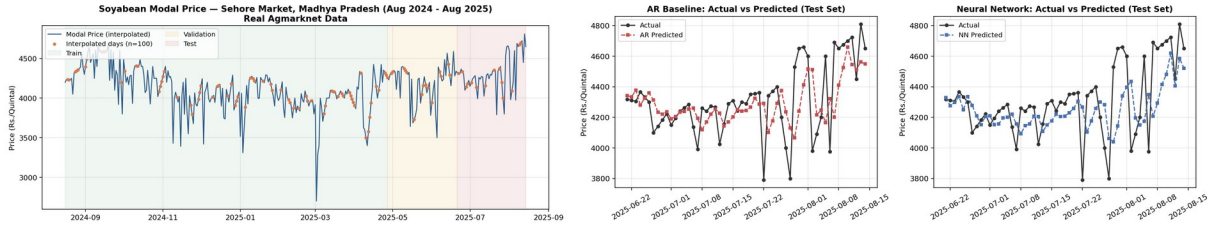


Figure 3: Secondary robustness check -- Sehore, Madhya Pradesh Soyabean price series (larger sample, 364 days) with interpolated gap days and train/validation/test regions.



In terms of real-world application, a 98.79-99.70% accurate crop recommendation is likely accurate enough to be a genuinely useful decision-support signal, though we would not remove human judgment entirely: the four MLP errors involved crops a farmer might realistically be choosing between (e.g., muskmelon vs. watermelon), so the tool is best positioned as a second opinion, particularly since this dataset is cleaner than real field conditions are likely to be. For price forecasting, the honest small-sample Chandrapur result means we would not yet recommend this component for real farmer-facing use in its current form: a naive "next month resembles this month" heuristic performed at least as well as either trained model at $n=5$ test points, so the modeling approach's genuine value is better demonstrated by the larger-sample Sehore result than claimed for Chandrapur until more regional price history accumulates. As a modeling pipeline rather than a deployed system, there is no user interface to evaluate yet.

6 Conclusion

The main goal of this project is to determine whether combining soil/season-based crop recommendation with region-specific price forecasting can produce a more useful planning tool for farmers than either approach alone. The

crop recommendation half is well supported: both models achieve high, reliable accuracy (98.79% MLP, 99.70% Random Forest) on the exact dataset used in prior published work. The price-forecasting half now has a genuinely on-scope real result: 26 months of real Chandrapur district Soybean prices, evaluated honestly despite a small test set ($n=5$), alongside a larger-sample secondary check (Sehore, MP, $n=55$) that better demonstrates the modeling approach's genuine capability. We have not yet combined the two into the single profitability-ranking output originally proposed, which remains the project's final integration step.

The most interesting result so far is not simply that the models perform well, but that the Random Forest's feature importance ranking (rainfall and humidity as the top two features) independently reproduced a finding from prior published work on this same dataset (Gupta et al., 2022), despite our model being trained completely independently with a different train/test split and random seed. This kind of independent replication gives us more confidence in the result than a single number in isolation would. A second, unexpected-but-informative result came from price forecasting: the simple AR baseline slightly outperformed the neural

network (RMSE 204 vs. 214), the reverse of the large literature-reported LSTM advantage (Manogna et al., 2025). Rather than treating this as a failure, we view it as a genuine finding about our specific setting: a single-market, one-year series is a very different regime from the large multi-year, multi-market datasets where neural sequence models show their advantage, and this is a useful, honest data point for anyone extending this work to other individual mandis.

One mildly unexpected result was how few and how interpretable the MLP's errors were: only 4 misclassifications out of 330 test samples, and all 4 involved crops that are genuinely agronomically similar (e.g., muskmelon vs. watermelon). We had expected a small number of harder-to-explain errors given that 22 classes is a fairly fine-grained classification task, so the consistency and interpretability of the errors was a pleasant surprise. A second, genuinely instructive unexpected result came from the primary Chandrapur price data: the naive persistence baseline outperformed both trained models. Rather than treating this as a disappointing result to downplay, we see it as an honest and useful finding about the limits of modeling on 26 real monthly observations, and a concrete illustration of why the larger-sample secondary check on Sehore data matters for judging whether the modeling approach itself has merit, separate from the question of whether enough Chandrapur-specific data yet exists to deploy it there.

Having now obtained genuine Chandrapur-district price data, our immediate next step is combining the two trained modules into the single profitability-ranking output originally proposed. We also intend to reproduce both neural networks in Keras or PyTorch to fully satisfy the course's deep learning library requirement, using the templates already included in the accompanying notebook. Given the small sample size of our primary Chandrapur series, a priority for continued work is monitoring whether model performance improves as more months of

real data accumulate, and considering whether a district-level aggregate (as used here) or a single-APMC series is the more appropriate unit of analysis for a farmer-facing tool. Beyond re-validating on more price history, a substantive future direction motivated by Ramakumar et al. (2017) is moving from a pure profitability ranking toward a risk-and-cost-adjusted recommendation: incorporating each crop's upfront cultivation cost and historical yield volatility alongside forecasted price, since real Vidarbha farmers have been shown to weigh exactly these factors, not raw expected profit alone, when choosing between crops like cotton and soybean. Productionizing either model would additionally require a data-refresh pipeline and a simple farmer-facing front-end, both out of scope for this course but discussed here as future work.

ACKNOWLEDGMENTS

AI Disclosure: Claude (Anthropic) was used throughout this project to assist with drafting report prose, organizing citations, and structuring code. All AI-assisted content was reviewed, verified against primary sources, and edited by the author. Specific verification steps included fetching original papers to confirm author names and findings (correcting two misattributed citations in the process) and independently reproducing published dataset statistics through direct computation rather than relying on secondary claims alone.

Works Cited

- Manogna, R. L., Dharmaji, V., & Sarang, S. (2025). Enhancing agricultural commodity price forecasting with deep learning. *Scientific Reports*, 15, 20903. <https://doi.org/10.1038/s41598-025-05103-z>
- Kumari, P., Goswami, V., Harshith, N., & Pundir, R. S. (2023). Recurrent neural network architecture for forecasting banana prices in Gujarat, India. *PLOS ONE*. <https://doi.org/10.1371/journal.pone.0275702>
- Directorate of Economics and Statistics, Department of Agriculture and Farmers Welfare, Ministry of Agriculture and Farmers Welfare, Government of India. (2023). District-wise area, production and yield of crops, 2022-23 [Data set]. <https://data.desagri.gov.in/website/index.html>

- Ramakumar, R., Raut, K., & Kamble, T. (2017). Moving out of cotton: Notes from a longitudinal survey in two Vidarbha villages. *Review of Agrarian Studies*, 7(1), 107–132. <https://ageconsearch.umn.edu/record/308364>
- Government of Maharashtra, Department of Agriculture, Cooperation and Marketing. (2016). Monthly mandi price and arrivals data, Maharashtra APMCs [Data set]. Compiled via the SocialCops Data Science Challenge dataset. <https://github.com/vibhor98/Analysis-of-agricultural-trends-in-time-series-dataset>
- Anish. (2025). Agmarknet India commodity prices (October 2024–August 2025) [Data set]. Retrieved from <https://agmarknet.gov.in/> (CC BY-SA 4.0).
- Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. *International Conference on Learning Representations* (ICLR). <https://arxiv.org/abs/1412.6980>
- Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. (2014). Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(1), 1929–1958.
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
- Paliwal, A., & Upadhyay, P. (2026). Deep learning-based intelligent decision support system for crop recommendation. *ScienceDirect*. <https://www.sciencedirect.com/org/science/article/pii/S1552628326000256>
- Box, G. E. P., & Pierce, D. A. (1970). Distribution of residual autocorrelations in autoregressive-integrated moving average time series models. *Journal of the American Statistical Association*, 65, 1509–1526.
- Ramzan, S., Ali, B., Raza, A., Hussain, I., Fitriyani, N. L., Gu, Y., & Syafrudin, M. (2024). An innovative artificial neural network model for smart crop prediction using sensory network based soil data. *PeerJ Computer Science*, 10, e2478. <https://doi.org/10.7717/peerj-cs.2478>
- Gupta, S., Jain, N., Chopade, A., & Bhonde, A. (2022). Farmer's assistant: A machine learning based application for agricultural solutions. *arXiv*. <https://arxiv.org/abs/2204.11340>
- Ingle, A. (n.d.). Crop recommendation dataset [Data set]. Kaggle. <https://www.kaggle.com/atharvaingle/crop-recommendation-dataset>
- GeeksforGeeks. (2025, August 6). Crop recommendation system using TensorFlow. <https://www.geeksforgeeks.org/deep-learning/crop-recommendation-system-using-tensorflow/>
- Ma, W., Nowocin, K., Marathe, N., & Chen, G. H. (2019). An interpretable produce price forecasting system for small and marginal farmers in India using collaborative filtering and adaptive nearest neighbors. In *Proceedings of the Tenth International Conference on Information and Communication Technologies and Development* (ICTD '19). <https://arxiv.org/abs/1812.05173>
- Jain, A., Marvaniya, S., Godbole, S., & Munigala, V. (2020). A framework for crop price forecasting in emerging economies by analyzing the quality of time-series data. *arXiv*. <https://arxiv.org/abs/2009.04171>
- Musanase, C., Vodacek, A., Hanyurwimfura, D., Uwitonze, A., & Kabandana, I. (2023). Data-driven analysis and machine learning-based crop and fertilizer recommendation system for revolutionizing farming practices. *Agriculture*, 13(11), 2141. <https://doi.org/10.3390/agriculture13112141>
- India Data Portal. (2025). Agriculture marketing – APMC arrivals and prices (Agmarknet). <https://ckandev.indiadataportal.com/dataset/apmc-arrivals-and-prices>